Online Supplement

Daughters' Days: Son Preference, Gender Norms and the Time Use of Sisters and Brothers in India

Replication in the 2019 survey

With one harmonised specification in both rounds (children of the head aged 10 to 17 in households with at least one girl and one boy in that group, household, age and diary-day fixed effects), girls do 63.9 minutes (SE 1.8) more unpaid domestic work than their brothers in 2019 (20,253 children) and 58.1 minutes (SE 1.4) in 2024 (19,379 children) (National Statistical Office, 2020; National Statistics Office, 2025). This specification differs from the main sibling sample, so the levels are not comparable to the main estimates; the point is that a large within-household gap appears in both rounds, somewhat smaller in 2024.

Table S1: From the household roster to the analysed diaries

Step	N
Persons aged 6+ on the household roster	454,192
Without diary records in the file	3,735
Diaries (person-days)	450,457
Diaries not summing to exactly 1,440 minutes	5
Third-gender diaries (excluded from women-men comparisons)	49
Diaries of women and men	450,408

Notes: All counts are computed from the microdata at knit time. Persons without diary records appear only on the household roster.

Table S2: Activity categories and ICATUS 2016 codes

Category	ICATUS 2016 codes
Employment	110, 121–129, 131–136, 139, 141–142, 150, 160, 170
Employment-related travel	181–182
Own-use production of goods	211–218, 221–227, 229–230, 241–242, 250
Food and meals	311–314, 319
Cleaning and upkeep	321–325, 329
Care of textiles	341–344, 349
Shopping	371–372
Unpaid domestic services (division 3)	311–314, 319, 321–325, 329, 331–333, 339, 341–344, 349, 351–352, 359, 361–362, 369, 371–372, 380, 390
Childcare	411–417, 419, 442
Care of dependent adults	421–426, 429, 443
Unpaid caregiving (division 4)	411–417, 419, 421–426, 429, 431–432, 439, 441–444, 490
Learning	611–614, 619–620, 630, 640, 690
Free time (divisions 7 and 8)	711–713, 719, 721–723, 729–730, 741–742, 749–750, 790, 811–813, 819, 821–823, 829, 831–832, 841–843, 849–850, 860, 890
Sleep	911–913, 919

Notes: Minutes are summed over major activities in the person-day diary. Codes are three-digit ICATUS 2016 codes as recorded in the survey.

0.1 Mothers' time and the sex composition of children

The sibling comparison holds the parents fixed, so it cannot show whether parents' own time depends on the sex of their children. Table S5 compares mothers whose co-resident children aged 6 to 17 are all daughters with mothers who have at least one such son, conditional on the number of children and household characteristics. Mothers of only daughters do 2.7 minutes (SE 0.8) more

Table S3: Construction of the analysis samples

Step	N
Diaries in the file	450,457
Diaries of women and men (third-gender diaries excluded)	450,408
Children aged 6–17, never married, children or grandchildren of the head	82,223
In a set with at least two children aged 6–17	54,484
In a mixed-sex set (fixed-effects sample)	34,133
Mixed-sex sets	13,984
Mixed-sex sets with a unique eldest child	13,825
Children with a linked mother	31,832
Linked mothers aged 20–49 (mothers' analysis)	45,732

Notes: A sibling set is the children with the same relationship to the head (child or grandchild) in the same household.

Table S4: Construction and validation of the person-day panel

Check	Value
Activity records	10,211,478
Person-days (diaries)	450,457
Households	139,487
First Stage Units	9,969
Diaries summing to 1,440 minutes	100.0%
Records flagged minor (secondary)	17.5%
Persons with varying weight	0.00%
Invalid activity codes	None
Married persons with linked spouse	88.5%
Under-6 count method	direct count (file contains members under 6)

Notes: All activity records are used; no row sampling.

Table S5: Mothers' time and the sex composition of their co-resident children

Mother's minutes per day	All daughters (6–17)	Daughters 10–17 (vs sons)	Mean
Active childcare	2.7*** (0.8)	1.3** (0.5)	54.4
All childcare	3.2*** (0.8)	1.7*** (0.5)	56.9
Unpaid domestic work	-7.7*** (1.4)	-9.8*** (1.2)	365.6
Employment	5.3** (2.1)	8.8*** (1.8)	102.6

Notes: Linked mothers aged 20–49 with at least one co-resident child aged 6–17 (N = 45,732). Column 1: indicator that all such children are daughters. Column 2: number of daughters aged 10–17, controlling for the number of children aged 10–17 (so the coefficient compares a daughter with a son). Controls: mother's age band and education, urban, consumption quintile, number of children aged 6–17, members under six, employment (except for the employment outcome), proxy report, and state and day-of-week fixed effects. FSU-clustered standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

active childcare and 3.2 minutes (SE 0.8) more childcare in total. Their domestic work differs by -7.7 minutes (SE 1.4) and their employment time by 5.3 minutes (SE 2.1).

The second column asks whether daughters substitute for their mothers' domestic work. Condi-

tional on the number of children aged 10 to 17, each adolescent daughter in place of an adolescent son is associated with 9.8 minutes (SE 1.2) less domestic work by the mother. Mothers with adolescent daughters do less domestic work, which is consistent with daughters taking over part of their mothers' load.

These estimates lack a household fixed effect. Under son-biased stopping, mothers with only daughters differ from others in ways that may be related to their time use: they may have more children, may be more likely to be still having children, and may be in families with weaker or stronger son preference. I control for the number of children, the mother's age and household characteristics, but treat these results as secondary evidence.

0.2 Heterogeneity

Table S6: Heterogeneity of the girl-boy gap: minutes per day

	Domestic	Care	Learning	Free time
Urban household (N = 34,133)				
Girl	44.6*** (1.1)	1.8*** (0.3)	0.5 (1.7)	-35.5*** (1.4)
Girl × moderator	-16.1*** (2.2)	-0.8* (0.5)	9.2*** (3.5)	7.3** (3.0)
Top two consumption quintiles (N = 34,133)				
Girl	43.8*** (1.1)	2.1*** (0.3)	2.1 (1.7)	-35.2*** (1.4)
Girl × moderator	-11.1*** (2.0)	-1.9*** (0.4)	1.9 (3.3)	5.2* (2.9)
Mother has secondary education or more (N = 31,832)				
Girl	48.8*** (1.2)	1.7*** (0.3)	0.2 (1.9)	-37.0*** (1.5)
Girl × moderator	-32.2*** (1.6)	-1.1*** (0.4)	12.4*** (2.9)	11.8*** (2.5)
Head has secondary education or more (N = 33,911)				
Girl	47.5*** (1.2)	1.8*** (0.3)	0.4 (1.9)	-36.6*** (1.5)
Girl × moderator	-22.8*** (1.7)	-0.8** (0.4)	8.0*** (2.8)	9.4*** (2.6)
Child under six in household (N = 34,133)				
Girl	42.5*** (1.1)	0.1 (0.2)	4.1** (1.7)	-35.2*** (1.4)
Girl × moderator	-6.8*** (1.9)	6.4*** (0.7)	-6.3** (3.2)	5.8** (2.7)
Diary day is a Sunday (N = 34,133)				
Girl	38.7*** (1.0)	1.5*** (0.2)	1.6 (1.6)	-31.1*** (1.3)
Girl × moderator	22.7*** (3.3)	1.0 (1.1)	11.0*** (3.6)	-28.0*** (4.5)

Notes: Each block and column is one regression on mixed-sex sets with sibling-set fixed effects, single-year age effects, girl, the moderator and girl x moderator. Girl is the gap where the moderator is zero. The moderators other than the diary day are constant within the set. Mother's education is available where a unique mother is linked. FSU-clustered standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table S6 reports how the gap varies with household characteristics. The domestic-work gap in urban households differs from that in rural households by -16.1 minutes (SE 2.2), and in the top two consumption quintiles from that in the bottom three by -11.1 minutes (SE 2.0). The gap is present in urban and in richer households, so it is not only a response to poverty or to the demands of rural household production. In this respect it differs from the excess mortality of girls documented by Rose (1999), which responds to income shocks.

The gap is 48.8 minutes (SE 1.2) in sets whose mother has less than secondary education, and the difference for mothers with secondary education or more is -32.2 minutes (SE 1.6). For the household head's education the corresponding difference is -22.8 minutes (SE 1.7). Mothers' education is

associated with a smaller division of domestic work between their daughters and sons, consistent with the transmission of attitudes from mothers to children documented by Dhar et al. (2019).

Where a child under six lives in the household, the caregiving gap changes by 6.4 minutes (SE 0.7). Older girls are drawn into the care of young children when there are young children to care for, which is the sibling-care channel through which a mother's fertility can affect her daughters' time.